

Electromagnetic Physics

Sem 2

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CATEGORY: PYQs

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[Page 1]

Reg No. A2 o 317000 oj2 B.Tech/ M. Tech Integrated DEGREE EXAMINATION, JANUARY 2023 First Semester 21PYB101J-PHYSICS: ELECTROMAGNETIC THEORY, QUANTUM MECHANICS, WAVES AND OPTICS (For the candidates admitted from the academic year 2022-2023) Note: Part A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40 minute. Part-B and Part -C should be answered in answer bookict.) (i) Max. Marks: 75 Time: 3 Hours Marke B CO PO PART-A (20 x1 -20Marks) Answer ALLL Questions 1. If divergence of certain vector field is zero at any point. Then the vector field is (A) Non-conservative (C) Irotational (B) Rotational (D) Solenoidai 2. is defincd as the ratio of the permittivity of the medun and permittivity of free space. (A) Dielectric constant (C) Polarizability (B) Polarization vector (D) Electronic polarization 3 theorem related the surface integral of the curl of a vector to the line integral of the vector. (A) Stokes (C) Greens (B) Gauss divergetce (D) Gaus theorern 4. If the charge is spread out along a line with charge per unit length 2 (λ), the dq_ (A) $dq = \lambda dl$ (C) $dq = \lambda dt$ (B) $dq = \lambda da$ (D) $dq = \lambda dr$ 2 2 5. Ferrites are the modified structure of (A) Cobalt (C) Nickel (B)ron (D) Gold 2 2 6. In geneal the chemical formula of a fermite molecule is (A) MFeo (C)MFeo (B) M Feo (D) MFeO 7. A tuny movable magnetized cylindrica! volume in thin magnetic material is 2 2 called (A) Ferrite (C) Magmetic bubble (B) Gamet (D) Magneto plumbite P of

[Page 2]

8. Magnetoplumbites belong to a family of_ (A) Ferrite (C) Magnetic thin film 1 (B) Magnetic bubble (D) Garnet 9. A perfect blackbody is said to be a system which wavelengths of radiation. (A) Transmit C) Diffracts all possible 1 2 (B) Absorbs and Emits (D) Creates photon 12 3 nature of matter statistically. (B) Temperature and photon (D) Pressure and wave 10. Wave function relates the and (A) Photon and pressure (C) Particle and wave 31 effect is a phenomenon in which electrons are ejected from i1 3 11. The the surface of a metal when light is incident on it. (A) Photoelectric (C)Raman (B) Compton D) Thomson 1 12. The Bragg's diffraction law is (A) $d \sin \theta = n \lambda$ (C) $2d \cos \theta = n \lambda$ 231 (B) $2d \sin \theta = n \lambda$ (D) $2d \tan \theta = n \lambda$ 1 41 13. How many lenses are used in Fresnel diffraction? (A) Two convex lenses (C) One convex lens (B) Two concave lenses (D) No lens used 11 4 2 14. The main principle used in interference is (A) Heisenberg's principle (C) Quantum mechanics uncertainty (B) Superposition principle (D) Fermi principle 15. Light waves are

transverse in nature, can be demonstrated by observing the phenomenon of_ (A) Interference (C) Polarization

141 (B) Diffraction D) Double reflection 16. According to Brewster' s law, the refractive index of transparent medium is given by (A) $\mu = \cos i$, 11 4 1 (B) $\mu = \sin i$, (C) $\mu = \cot i$ D) $\mu = \tan i$, 17. What is the condition of population inversion achieved in laser action? (A) $N > N_1$ N= N_1

1151 B) $N_a < N$ (D) $N_a = N = 0$ 18. A pair of mirrorS placed on either side of the active medium is known as 5 (A) Cavity (C) Optical resonator (B) Case D) Optical pumping Page 2 of 4 5YF121PYR1813

[Page 3]

19, Which of the following is not a characteristic of lasers? (A) Monochromatic (C) Divergent 115 B) Coherent D) Intense 20. How does the refiactive index vary in graded index fibre? (A) Tangentially (C) Longitudinally 2 5 1 (B) Radially D) Transversely PART B (5 x8 = 40 Marks) Answer ALL Questions Marks BL CO P0o

21. a.i. Apply Gauss law to find the electric field intensity due to a line charge of 3 infinite length. 3 1 1 3 i. A point charge of 13.5×10^{-5} C is enclosed at the centre of the cube of side 6.0 cm. Find the electric flux through the whole volume and thorough the face of the cube. (OR) b. Analyze the various polarization mechanisms in dielectric material. 8 411

5 42 22. ai. Distinguish soft and hard magnetic material based on magnetic properties of materials. 3 3 2 1 ii. A magnetic field of 1800 ampere metre produces a magnetic flux of 3×10 weber in an iron bar of cross sectional area 0.2 cm^2 . Calculate permeability. (OR) 3 2 b. Describe the mechanism of giant magneto resistance and tunnel magneto resistance. 83 3 1 23. a. Apply uncertainty principle to analyze the non-existence of electrons in the nucleus. (OR) 8 33 1 b. Describe the experimental verification of Davisson and Germer's difiraction experiment. 24 11 24. a. Describe the Fraunhofer difiraction due to double slit with neat sketch. (OR) 41 b. Analyse the production and detection of circularly poiarized light. 25 1 25. a. Discuss the essential components of laser system and pumping mechanisms. (OR) b. Obtain the expression for numerical aperature and acceptance angle for an optical fiber. eSJF121PYB1013 Page 3 of 4

[Page 4]

Marks BL CO PO PART C (1x15-15 Marks) Answer ANY ONE Questions

4 26.i. Analyze the Maxwell's equations in free space. 4 11 ii. Calculate the characteristic impedance of the medium. 15 4 31 27. A particle moving ina one-dimensional box is described by $V=0$ for $0 < x < L$ for $0 \leq x \leq 2L$ Write and solve its Schrodinger's wave equation, obtain the Eigen value and Eigen function and find the normalization of the wave function.

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[Page 1]

b. Explain the production and detection of circularly polarized light using quarter wave plate?

32. a. Explain the modes of vibrations of CO₂ molecule. Describe the construction and working of

CO₂ Laser with necessary diagrams.

(or)

b.i. Define Numerical aperture and derive an expression for numerical aperture? (8 Marks)

ii. Find the relative population of the two states in a Nd-YAG laser that produces a light beam

of wavelength 6943 Å at 300 K. (4 Marks)

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Reg. No.

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018

First Semester

18PYB101J - PHYSICS: ELECTROMAGNETIC THEORY, QUANTUM MECHANICS,
WAVES AND OPTICS

(For the candidates admitted during the academic year 2018-2019)

Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed

over to hall invigilator at the end of 45th minute.

Part - B and Part - C should be answered in answer booklet

Note:

(r)

(ir)

Time: Three Hours Max. Marks: 100

PART-A(20x1=20Marks)

Answer ALL Questions

1. is defined as the number of magnetic lines of force passing through an

2. The vector field whose divergence is zero is called

(A) Irrotational

(C) Conservative

(B) Rotational

3. The dipole moment per unit volume of the dielectric material is called

4. Orientation polarization arises due to the presence of

(A) Conductors

(C). Semiconductors

(B) Solenoidal

(D) Polarization vector

(A) Dielectric constant

(B) Polar molecules

(D) Superconductors

unit area of cross section.

- (A) Magnetic flux density
- (C) Magnetic field intensity
- (A) Polarizability
- (C) Permittivity
- (A) Very steep
- (C) Negligible
- (A) Giant magnetoresistance
- (C) Clossal Magnetoresistance

- (A) Potential wall
- (C) Magrretic wall
- @) Magnetic flux
- @) Intensity of magnetization

5. In soft magnetic materials, the nature of the hysteresis loop is

- @) Very broad
- (D) Straight line

thin magnetic material ts6. A tiny movable magnetiznd cylindrical volume in called

- (A) Garnet @) Magnetoplumbites
- (C) Magnetic bubble (D) Ferrites

7. Two ferromagnets are separated by a few atomic layers of insulator are known as -----
--

8. The boundary walt benveen domains is known as

- (B) Tunnel magnetoresistance
- (D) Superconductor
- @) Blochwall
- (D) Semiconductor wall

17NF1I8PYB101JPage 4 of4 1711FU8PYn101J Page I of4

[Page 2]

9. The characteristics of wave function ry are

- (A) Real function" finite and (B) discontinuous
- (C) Complex, inflnite and discontinuous @)
- Complex, single value4 furite
- continuous function

Complex single vahred and infinite

20. Ina fiber, the refractive index changes in a step fashion from the centre

- @) .Graded index
- @) Glass
- of the fiber, to cladding.

- (A) Step Index
- (C) Photo

10.

frrnction

The existence of matter waves was experimentally proved by

(A) Raman

(C) De-Broglie

A variable quantity which characterizes de-Broglie waves is known as

-ffif;f,Sx.ffi'8ffi',Hl

Derive Poisson's equations.

Compare soft and hard magnetic materials.

Explain Blackbody Radiation with neat diagram.

Derive the expression for de Broglie wave length in terms of energy and voltage.

Write the differences between Fresnel's and Fraunhofer's diffraction.

Explain fiber optic communication system with a neat block diagram.

Derive Einstein's relations and hence deduce the expressions for the ratio of spontaneous emission rate to be stimulated emission rate.

PART-C (5 x 12=60Marks)

Answer ALL Questions

(B) Davisson and Germer

(D) Fresnel 2t.

11

11.

23.

24.

25.

26.

27.

(A) Photon

(C) Phonon

(B) Wave Function

@) Field

t2. is the probability of finding the particle inside the box.

@) Normalisation

(D) Interference

(B) $p: \cos^4$,

(D) $p: \cot^2$,

14. Nicol prism is made from Crystal

13.

(A) Quantisation

(C) Hybridisation

(A) $\mu_r = \sin \theta$,

(C) $\mu_p = \tan \theta$

Brewster's law in terms of refractive index can be expressed as

Calcite

15. In Fraunhofer diffraction the incident wavefront should be

@) Plane

@) Cylindrical

at infinite @) Source of light is kept at finite distance

from the aperture

(D) Concave lens is used

(B) Normal

(C) Dense

oxygen atoms oscillate along the axis of the molecule

dielectrics?

(or)

b.i. Derive an expression for Clausius-Mossotti equation. (8 Marks)

ii. Apply Gauss law to find the electric field intensity outside a uniformly charged spherical

(4 Marks)

29. a. What are ferrites? Explain their different types of structures. Write their applications.

(or)

b.i. Explain in detail about the theory of magnetic domains in ferromagnetic material. (8 Marks)

ii. Write a note on magnetoplasmas.

17. The minimum population inversion density required to overcome the losses is called population inversion.

(A) Elliptical

(C) Spherical

16. In Fresnel Diffraction

(A) Source of light is kept

distance from the aperture

(C) Convex lens is used

(A) Threshold

(C) Standard

18. In mode, the

(A) Symmetric stretching

(C) Bending

(A) Numerical Aperture

(C) Efficiency

(B) Asymmetric stretching

(C) Normal

(B) Cone

(C) Aperture

(4 Marks)

simultaneously departing or approaching the carbon atom which is stationary.

of the fiber is the light collecting efficiency of the fiber and is a measure

of the amount of light rays that can be accepted by the fiber.

30. a- Discuss the application of Schrodinger's wave equation to a particle enclosed in an one

dimensional potential box.

b.i. Describe the experimental verification of Bragg's law, and Germer's diffraction experiment.

(8 Marks)

ii. A neutron of mass 1.675×10^{-27} kg is moving with a kinetic energy 10 keV. Calculate the de-Broglie wavelength associated with it. (a Marks)

31. a. Describe the intensity distribution in Fraunhofer diffraction pattern due to a single slit?

19.

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CATEGORY: PYQs

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[Page 1]

IMPORTANT QUESTIONS

MODULE-1

4 Marks

1. Define Grad, Div and Curl and explain their physical significance
2. Explain Gauss-Divergence theorem and its significance
3. Explain Stoke's theorem and its significance
4. Explain Electric field and Electric potential due to point charge distribution
5. Explain Electric field and Electric potential due to line charge distribution
6. Explain Electric field and Electric potential due to surface charge distribution
7. Explain Electric field and Electric potential due to volume charge distribution
8. Using Gauss' law, find the electric field intensity due to line charge of infinite length
9. Derive the expression of Poisson's equation for electrostatic potential
10. Derive expression of Laplace equation for electrostatic potential
11. Derive the expression $\mathbf{J} = \nabla \times \mathbf{A}$
12. Derive the expression $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$
13. Derive continuity equation
14. State and explain Gauss's law
15. State and explain Biot-Savart law
16. State and explain Faraday's law
17. State and explain Ampere circuital law
18. Derive relationship between polarizability, permittivity and dielectric constant
19. Distinguish between polar and non-polar dielectrics

[Page 2]

8 Marks

1. Apply Gauss' law to find the electric field intensity inside and outside a uniformly charged spherical shell
2. Obtain Maxwell's equation for electromagnetism from fundamental laws of electricity and magnetism
3. Explain the various polarization mechanisms in dielectric
4. Describe the frequency and temperature dependence on polarization of dielectrics
5. Derive an expression for the internal field in a dielectric
6. Derive Clausius-Mossotti equation

MODULE-2

4 Marks

1. Compare Soft and Hard magnetic materials
2. Explain Hysteresis loop and energy product with diagram
3. Explain the concept of ferromagnetism
4. Write a note on Giant Magneto Resistance (GMR) with neat diagram
5. Write a note on Colossal Magneto Resistance (CMR) with neat diagram
6. Write a note on Garnets and its applications
7. What are Magnetoplumbites and give its applications
8. What are Multiferroic materials and explain its applications

8 Marks

1. Explain in detail about the theory of magnetic domain in ferromagnetic materials with neat diagram
2. What are ferrites? Describe the regular and inverse Spinel structure of ferrites with neat diagram. Write their applications

[Page 3]

3. What is magnetic bubble memory? Explain the principle and working of MBM and write their uses
4. Explain the properties, magnetic mechanism and processing of magnetic thin films.
5. Explain the function of Tunnel Magneto-resistance with neat sketch

MODULE-3

4 Marks

1. Write a note on inadequacies of classical mechanics
2. Explain photoelectric effect with neat diagram
3. Explain Compton effect with necessary diagram
4. What is blackbody radiation and explain Planck's theory of blackbody radiation?
5. Derive de-Broglie equation in terms of Energy/Voltage/Temperature
6. Heisenberg's uncertainty principle
7. Physical significance of the wave function

8. Normalization of wave function
9. Describe Davisson and Germer's diffraction experiment for the verification of matter waves or diffraction

8 Marks

1. Derive Schrödinger time independent equation
2. Derive Schrodinger time dependent equation
3. Application of Schrodinger equation for a particle in box (12)

[Page 4]

4. Linear Harmonic Oscillator- Energy and Wave function (12)
5. Hydrogen atom problem - radial, angular equation derivation and its solutions

MODULE-4

4 Marks

1. What is meant by diffraction of light?
2. Distinguish between interference and diffraction
3. Distinguish between Fresnel and Fraunhofer diffraction
4. Dispersive power of a plane diffraction grating
5. Resolving power of a plane diffraction grating
6. What is meant by plane polarized, circularly polarized and elliptical polarized light?
7. Distinguish between unpolarized and polarized light
8. How will you obtain plane polarized light by reflection?
9. Explain the terms plane of polarization and plane of vibration
10. Explain the phenomena of double refraction in a calcite crystal
11. Explain Brewster's law
12. State and explain law of Malus
13. Explain circular polarization
14. Explain elliptical polarization
15. Write a note on Quarter wave plate
16. Write a note on Half wave plate
17. State the laws of rotatory polarization

[Page 5]

8 Marks

1. Derive an expression for the intensity distribution due to Fraunhofer diffraction at a single slit.
2. Describe the Fraunhofer diffraction due to double slit
3. Give the construction and theory of plane transmission grating and explain the formation of spectra by it
4. Give the theory of plane transmission grating and show how would you use it to determine the wavelength of light
5. What do you understand by the resolving power of a grating? Derive the necessary expression
6. How will you produce and detect plane, elliptically and circularly

polarized light?

7. Show that the plane polarized and circularly polarized lights are the special cases of elliptically polarized light
8. Describe the construction of a Nicol prism and show how it can be used as a polarizer and as an analyzer.
9. Give the construction and theory of (i) quarter wave plate (ii) half-wave plate.
10. Give three methods for producing plane polarized light
11. Explain Brewster law and show that when light is incident on a transparent substance at the polarizing angle, the reflected and refracted rays are at right angles
12. State and explain Fresnel's relation at a dielectric interface

[Page 6]

MODULE-5

4 Marks

1. Characteristics of LASER
2. Explain Absorption and Emission process
3. Explain different pumping mechanisms
4. Essential components of a laser system
5. Distinguish between Step index and Graded index fiber
6. Distinguish between single mode and multimode fiber
7. Pressure sensor (intrinsic)
8. Position sensor (extrinsic)

8 Marks

1. Derivation of Einstein's co-efficient (8)
2. Carbon-di-oxide laser (3+5)
3. Nd:YAG Laser (2+6)
4. Semi-Conductor (Homo-junction) LASER (2+6)
5. Semi-Conductor (Hetero-junction) LASER (2+6)
6. Derivation of Numerical Aperture and acceptance angle or propagation characteristics of light through optical fiber (2+6)
7. Explain different types of fiber based on materials, no. of modes and refractive index profile (4+4+4)
8. Explain different kind of losses associated with optical fibers – Attenuation and dispersion
9. Fiber optic communication system
10. Intrinsic and extrinsic sensor